

Confined field energy as inertia, and an exact Z_3 identity underlying the Koide charged-lepton mass relation (v3)*

Gustaf Kranck^{1,†}

¹*Vagus Health Ltd., Cambridge, United Kingdom*

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We develop, in fully explicit form, the proposition that inertial mass is confined field energy, and we connect it to an exact algebraic structure underlying the Koide charged-lepton mass relation. First, for the (1+1)-dimensional ϕ^4 kink we assemble a set of exact classical statements: the soliton of a theory containing no particle and no kink-mass parameter carries energy $E = \gamma M$ and momentum $P = \gamma M u$ with M equal to the trapped field energy; Newton’s second law and the equality of inertial and gravitational mass follow as identities rather than postulates; and the kink supports a single internal oscillation mode at $\omega_1 = (\sqrt{3}/2)m$, furnishing the particle with an intrinsic clock whose boost dilation is derived, not assumed. Second, we show that if the three charged-lepton square-root masses are the eigenvalues of a generation-symmetric (democratic plus Z_3 -circulant) coupling operator, $\sqrt{m_k} = \mu [1 + 2\kappa \cos(\delta + 2\pi k/3)]$, then the Koide ratio $Q \equiv \sum m_k / (\sum \sqrt{m_k})^2$ equals $2/3$ exactly if and only if $\kappa^2 = 1/2$, an equipartition condition between the democratic and symmetry-breaking sectors; geometrically, $Q = 2/3$ states that the vector $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes an angle of exactly 45° with the democratic axis. Confronting this with the current Particle Data Group world-average masses gives $Q = 0.6666645(51)$, consistent with $2/3$ at 0.4σ , and a fitted phase $\delta = 0.2222248(63)$, consistent with the rational value $2/9$ at 0.4σ . Under a structureless null hypothesis (masses drawn log-uniformly over the observed span), the probability of $|Q - 2/3|$ falling within the experimental band is $\approx 1.1 \times 10^{-5}$. The construction yields a sharply falsifiable corollary: for n generations coupled through a Z_n -circulant with equipartition, $Q_n = 2/n$. For $n = 4$ the condition $Q_4 = 1/2$ has, with the measured lepton masses, exactly two roots, $m_{\ell_4} = 95.1$ MeV and 9.32 GeV, both long excluded; the framework thus predicts unconditionally that no sequential fourth charged lepton exists. A four-loop $\overline{\text{MS}}$ analysis of the quark sector, executed here, shows that pure QCD leaves the circulant parameters exactly scale-invariant, and that the fitted amplitudes $\kappa_{\text{up}} = 0.9114(2)$ and $\kappa_{\text{down}} = 0.7880(7)$ both exceed $1/\sqrt{2}$ but are incompatible with a single parameter: equipartition is confined to the lepton sector as far as this test reaches. We conclude with a graded program of experimental tests, ranging from Belle II τ -mass precision—which sharpens the $\delta = 2/9$ test threefold—through hadronic breathing-mode spectroscopy of the soliton internal clock, to two tabletop Casimir-force configurations that probe, and at minimum bound, the coupling of vacuum structure to static and rotating vector-potential topology. All symbolic results are machine-verified and all numerical procedures are specified for reproduction.

I. INTRODUCTION

Most of the mass of ordinary matter is not a primitive attribute of constituents but confined field energy. Lattice computations determine the light-hadron spectrum *ab initio* from quantum chromodynamics (QCD) [1], and decompositions of the proton’s energy–momentum tensor attribute the overwhelming share of its mass to quark and gluon field energy rather than to Higgs-induced quark rest masses [2]; the gluon fields themselves are massless. In Wilczek’s phrase, this is “mass without mass” [3]: the dominant mass in the visible universe is the energy of massless gauge fields trapped in self-sustaining configurations.

This paper takes that fact as its starting point and pursues two questions in the most explicit setting available for each.

(i) *Mechanism.* If mass is confined field energy, then inertia—the resistance to acceleration quantified by the m in $F = ma$ —must be derivable from field dynamics alone, with no mass parameter for the composite object introduced by hand, and the equality of inertial and gravitational mass [4, 5] must emerge as an identity. Section II demonstrates this in the exactly solvable case: the (1+1)-dimensional ϕ^4 kink. The individual classical ingredients are long known [6–10]; our purpose is to assemble them into four sharp statements (inertia, equivalence, internal clock, topological protection) that together elevate “mass is trapped propagating energy” from an intuition to a theorem of a fully specified model, and to make each statement machine-verifiable. Section III recalls why the (3+1)-dimensional continuation of this mechanism is not hypothetical: Derrick’s obstruction [11] is evaded by known topological solitons, and the Skyrme description of baryons [12–14] together with the QCD mass budget [1, 2] realizes the mechanism in nature.

(ii) *Spectrum.* If elementary fermion masses are coupling depths to a common structure, the observed mass spectrum should reflect the symmetries of the coupling operator. Section IV analyzes the charged-lepton sector,

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† gustaf@vagus.co

where an empirical relation of striking precision has been known since 1981: the Koide ratio [15–17]

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (1)$$

holds within present experimental uncertainty. We show that Eq. (1) is an *exact identity* of any spectrum generated by a democratic-plus- Z_3 -circulant operator with a fixed perturbation-to-background amplitude $\kappa = 1/\sqrt{2}$, a condition we characterize as spectral equipartition; the known circulant parametrization [18–20] is thereby given an if-and-only-if characterization and a variational reading. We then confront the parametrization with the current Particle Data Group (PDG) world-average masses [23], propagate experimental uncertainties, quantify the coincidence probability under a structureless null hypothesis, and state a falsifiable generalization, $Q_n = 2/n$. Radiative stability of relations of this type is addressed by the Sumino mechanism [21, 22].

Section VI states explicitly which results are established mechanisms, which are reinterpretations of known physics, and which remain conjectural, and lists the calculations that would promote the latter. Throughout we use natural units $c = \hbar = 1$; all physical claims are carried by dimensionless quantities.

II. INERTIA FROM CONFINEMENT: EXACT SOLITON STATEMENTS

A. Model and kink

Consider a real scalar field in 1+1 dimensions,

$$\mathcal{L} = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}(\partial_x \phi)^2 - \frac{\lambda}{4}(\phi^2 - v^2)^2, \quad (2)$$

with equation of motion

$$\partial_t^2 \phi - \partial_x^2 \phi + \lambda \phi (\phi^2 - v^2) = 0. \quad (3)$$

The theory contains two parameters, λ and v . Small oscillations about either vacuum $\phi = \pm v$ have mass

$$m^2 = V''(v) = 2\lambda v^2, \quad (4)$$

and these quanta (“mesons”) are the ordinary propagating excitations. No other mass parameter appears anywhere in the theory.

Equation (3) admits the static kink [8]

$$\phi_K(x) = v \tanh\left(\frac{x}{W}\right), \quad W = \frac{1}{v} \sqrt{\frac{2}{\lambda}}, \quad (5)$$

which interpolates between the two vacua. Substitution of Eq. (5) into Eq. (3) gives an identically vanishing residual; we have verified this, and every symbolic statement

below, by computer algebra (Appendix A). The kink satisfies the Bogomolny identity $\frac{1}{2}(\phi'_K)^2 = V(\phi_K)$ pointwise [9, 10], and its energy is

$$E_0 = \int dx \left[\frac{1}{2}(\phi'_K)^2 + V(\phi_K) \right] = \frac{2\sqrt{2}}{3} \sqrt{\lambda} v^3 = \frac{m^3}{3\lambda}. \quad (6)$$

Two features deserve emphasis. First, E_0 is assembled entirely from λ and v ; the kink has no mass parameter of its own. Second, at weak coupling $E_0/m = m^2/3\lambda \gg 1$: the composite object is parametrically heavier than the field quantum. A mass hierarchy is generated by structure, not by input.

The kink carries the topological charge

$$Q_{\text{top}} = \frac{\phi(+\infty) - \phi(-\infty)}{2v} = \pm 1, \quad (7)$$

conserved not by Noether’s theorem but by topology: no continuous finite-energy evolution connects sectors of different Q_{top} . The object’s permanence is a conserved integer.

B. Exact relativistic energy and momentum

Because Eq. (3) is Lorentz covariant, the boosted profile

$$\phi(x, t) = v \tanh\left[\frac{\gamma(x - ut)}{W}\right], \quad \gamma = \frac{1}{\sqrt{1 - u^2}}, \quad (8)$$

is an exact solution (verified residual: zero). The moving kink is Lorentz-contracted by γ because it *is* field, and the field equation demands the contraction.

Let $\xi = \gamma(x - ut)$, so that $\partial_t \phi = -u\gamma \phi'_K(\xi)$, $\partial_x \phi = \gamma \phi'_K(\xi)$, and $dx = d\xi/\gamma$. Using the Bogomolny identity and $\int (\phi'_K)^2 d\xi = E_0$,

$$E(u) = \int dx T^{00} = \frac{1}{\gamma} \left[\frac{\gamma^2(1 + u^2)}{2} + \frac{1}{2} \right] E_0 = \gamma E_0, \quad (9)$$

$$P(u) = - \int dx \partial_t \phi \partial_x \phi = u \gamma E_0. \quad (10)$$

Both equalities are exact; the symbolic residuals $E - \gamma E_0$ and $P - \gamma E_0 u$ vanish identically, and a direct numerical quadrature at $(\lambda, v, u) = (2, 1, 0.6)$ reproduces $E = \gamma E_0 = 1.666667$ to the precision of the integration (Appendix A).

C. Newton’s second law and the equivalence principle as identities

Expanding Eq. (9),

$$E(u) = E_0 + \frac{1}{2} E_0 u^2 + \mathcal{O}(u^4). \quad (11)$$

The coefficient of $\frac{1}{2}u^2$ —the operational definition of inertial mass—is exactly the trapped field energy. Under a weak external perturbation, $dP/dt = F$ with Eq. (10) gives $F = Ma$ with $M = E_0$. Inertia is thereby exhibited as a mechanism: accelerating the kink requires Lorentz-deforming the entire trapped configuration, and the energy cost of that deformation per unit velocity is E_0 . The “resistance to acceleration” is the deformation cost of confined field structure.

Coupling the model minimally to weak-field gravity, the Newtonian potential of any localized configuration is sourced by $\int T^{00} dx$. For the kink this integral is E_0 —the same quantity that appears in Eqs. (9) and (10). There is no second parameter available for the gravitational mass to be:

$$m_{\text{grav}} = \int T^{00} dx = E_0 = m_{\text{inert}}. \quad (12)$$

Two qualifications fix the register of this statement. First, Eq. (12) is not a theorem of the present framework: it is inherited from minimal coupling, which sources gravity by the same energy–momentum tensor whose integral defines inertia. Any minimally coupled field theory satisfies it; the content here is only that the soliton *realizes* the identity with no mass parameter anywhere in the theory that could spoil it—the equality is exhibited, not derived anew. Second, the weak-field language must be read as a Newtonian analogy at the model’s dimensionality: in two spacetime dimensions the Einstein tensor vanishes identically, so there is no dynamical 1+1D general relativity for the kink to couple to; Eq. (12) states that the quantity which would source the Newtonian potential of the (3+1)-dimensional continuation (Sec. III) coincides with the inertial mass. Within any framework in which mass is confined field energy, the equivalence principle—tested at the 10^{-15} level [4, 5]—is then not a fine-tuning to be explained but an arithmetic identity to be expected.

D. The internal clock

Linear fluctuations $\phi = \phi_K + \eta(x)e^{-i\omega t}$ obey the Pöschl–Teller spectral problem [6, 8]

$$-\eta'' + \lambda(3\phi_K^2 - v^2)\eta = \omega^2\eta, \quad (13)$$

whose discrete spectrum consists of exactly two states: the translation zero mode $\omega_0 = 0$ and one internal “shape” mode at

$$\omega_1 = \frac{\sqrt{3}}{2} m. \quad (14)$$

Equation (14) is exact, not numerical: the fluctuation problem (13) is the Pöschl–Teller spectral problem, whose discrete spectrum is known in closed form [6, 8]—exactly two bound states, the translation zero mode $\omega^2 = 0$ and the shape mode $\omega^2 = \frac{3}{4}m^2$, below the continuum threshold $\omega^2 = m^2$. Our finite-difference

diagonalization ($N = 8000$; Appendix A) serves only as confirmation, reproducing $\omega_1/m = 0.866024$ against $\sqrt{3}/2 = 0.866025$.

The kink is therefore not featureless: it possesses an intrinsic, discrete oscillation—an internal clock. Because the boosted kink of Eq. (8) is the Lorentz map of the rest kink, the internal mode oscillates at ω_1/γ in laboratory time: the moving particle’s clock runs slow, *derived* from field dynamics rather than postulated kinematically. It is natural to identify the soliton’s proper time with the phase accumulated by its internal mode, $\tau = 2\pi N_{\text{cyc}}/\omega_1$; the experimental demonstration that a particle’s Compton frequency can serve directly as a clock [24] shows that this identification is operationally meaningful for physical matter. Conversely, a configuration with no trapped structure—a free wave of the medium—possesses no discrete mode and hence no internal clock; this is the model’s shadow of the vanishing proper time of null trajectories.

III. FROM 1+1 TO 3+1 DIMENSIONS

Derrick’s theorem forbids static, finite-energy scalar solitons in three spatial dimensions [11], so Sec. II cannot itself be the physical mechanism. It does not need to be. The obstruction is evaded by well-known routes—higher-derivative stabilization, gauge fields, topological terms—and nature demonstrably employs them. In the Skyrme description, baryons are topological solitons of the pion field whose mass and spin arise from the field texture [12–14]; in QCD itself, *ab initio* spectroscopy [1] and energy–momentum-tensor decompositions [2] establish that the proton is, in overwhelming proportion, confined energy of massless gluon fields and nearly massless quarks. The mechanism proved exactly in Sec. II—inertia as the deformation cost of confined field energy, equivalence as the identity of one integral, internal structure as the origin of proper time—is thus the solvable limit of the mechanism that generates most of the mass of visible matter. What the Higgs sector supplies is the remaining, elementary-fermion rest masses; it is to the structure of that sector that we now turn.

IV. THE CHARGED-LEPTON SPECTRUM: AN EXACT Z_3 IDENTITY

A. Circulant coupling and the identity

Suppose the three charged-lepton square-root masses are the eigenvalues of a single real coupling operator $\mathcal{M}^{1/2}$ acting on generation space, and suppose no generation is dynamically preferred, so that $\mathcal{M}^{1/2}$ commutes with the cyclic permutation $P : 1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. The most general such operator is a circulant,

$$\mathcal{M}^{1/2} = \mu [\mathbb{1} + \kappa (e^{i\delta} P + e^{-i\delta} P^\dagger)], \quad (15)$$

with eigenvalues

$$\sqrt{m_k} = \mu \left[1 + 2\kappa \cos\left(\delta + \frac{2\pi k}{3}\right) \right], \quad k = 0, 1, 2. \quad (16)$$

Using $\sum_k \cos(\delta + 2\pi k/3) = 0$ and $\sum_k \cos^2(\delta + 2\pi k/3) = 3/2$,

$$\sum_k \sqrt{m_k} = 3\mu, \quad \sum_k m_k = \mu^2 (3 + 6\kappa^2), \quad (17)$$

so that

$$Q = \frac{3 + 6\kappa^2}{9} = \frac{1 + 2\kappa^2}{3}. \quad (18)$$

Proposition 1. *For the spectrum (16), $Q = 2/3$ holds for all μ and all δ if and only if $\kappa^2 = 1/2$.*

The Koide relation is thus an exact identity of Z_3 -symmetric coupling at a specific perturbation amplitude—and only at that amplitude. The condition $\kappa^2 = 1/2$ states that the Z_3 -breaking perturbation carries exactly the same spectral power as the democratic background: $\text{tr}[\mu\kappa(e^{i\delta}P + \text{h.c.})]^2 = \text{tr}[\mu\mathbb{1}]^2 = 3\mu^2$. We refer to this as *spectral equipartition* between the symmetric and symmetry-breaking sectors of the coupling operator. The parametrization (16) itself has appeared in the Koide literature [18–20]; the contribution here is the if-and-only-if characterization (18) and its equipartition reading, which converts the question “why $Q = 2/3$?” into the sharper pair: why $\kappa = 1/\sqrt{2}$, and why the particular phase δ (Sec. IV B).

Geometrically, with $\mathbf{s} = (\sqrt{m_0}, \sqrt{m_1}, \sqrt{m_2})$ and democratic axis $\mathbf{n} = (1, 1, 1)/\sqrt{3}$,

$$\cos^2 \theta = \frac{(\mathbf{s} \cdot \mathbf{n})^2}{|\mathbf{s}|^2} = \frac{(\sum \sqrt{m_k})^2}{3 \sum m_k} = \frac{1}{3Q}, \quad (19)$$

so $Q = 2/3 \iff \theta = 45^\circ$ exactly [18]: the square-root-mass vector bisects the democratic axis and the hierarchy plane.

B. Confrontation with data

We use the PDG charged-lepton masses [23], $m_e = 0.51099895069$ MeV, $m_\mu = 105.6583755$ MeV, and the current world-average τ mass, $m_\tau = 1776.93 \pm 0.09$ MeV, which includes the Belle II measurement $m_\tau = 1777.09 \pm 0.14$ MeV [28] (the τ uncertainty dominates all error budgets below). Directly,

$$Q_{\text{exp}} = 0.6666645(51), \quad (20)$$

consistent with $2/3$ at 0.4σ .

Fitting Eq. (16) with $\kappa = 1/\sqrt{2}$ fixed (two parameters, μ and δ , for three masses; the identity absorbs the third degree of freedom) yields

$$\mu = 17.715809 \sqrt{\text{MeV}}, \quad \delta = 0.2222248(63), \quad (21)$$

TABLE I. Circulant reconstruction of the charged-lepton masses from $(\mu, \delta) = (17.715809 \sqrt{\text{MeV}}, 0.2222248)$ with $\kappa = 1/\sqrt{2}$ and assignment $k = (0, 1, 2) \rightarrow (\tau, e, \mu)$. The parameters are quoted rounded; table entries are computed at full fit precision. Residuals reflect the 0.4σ deviation of Q_{exp} from exactly $2/3$.

	reproduced (MeV)	PDG (MeV)	rel. dev.
m_τ	1776.9309	1776.93(9)	5.3×10^{-7}
m_μ	105.65752	105.6583755	8.1×10^{-6}
m_e	0.5109169	0.51099895	1.6×10^{-4}
Q	2/3 (exact)	0.6666645(51)	0.4σ
δ	2/9 (conj.)	0.2222248(63)	0.4σ

where the uncertainty is propagated from Δm_τ . The fitted value of δ depends on the fit objective at the 10^{-6} level: the absolute least-squares fit on the three square-root masses with $\kappa = 1/\sqrt{2}$ fixed (the Levenberg–Marquardt procedure of Appendix A) yields Eq. (21), $\delta = 0.2222248$; an exact fit to $(\sqrt{m_\tau}, \sqrt{m_\mu})$, with the identity absorbing the third mass, gives $\delta = 0.2222265$, and a relative least-squares fit over all three masses gives $\delta = 0.2222219$ —a spread of 4.6×10^{-6} , the same order as $|\delta - 2/9|$ itself. Under every convention $\delta - 2/9$ remains within $\sim 5 \times 10^{-6}$, so the consistency with the rational value at 0.4σ is convention-independent. (An unconstrained fit with κ free returns κ consistent with $1/\sqrt{2}$ at the same 0.4σ and the same δ , an independent confirmation that the data select spectral equipartition.) The fitted phase is consistent with the rational value

$$\delta = \frac{2}{9} = 0.222222 \dots \quad (22)$$

at 0.4σ —the same significance level at which Q is consistent with $2/3$; the two rational values stand or fall together. The proximity of δ to $2/9$ was observed numerically in Refs. [19, 20]; Eq. (21) sharpens it to a statement with a propagated experimental uncertainty. Table I summarizes the reconstruction.

C. Null-hypothesis comparison

How surprising is Eq. (20)? For any positive masses, $Q \in [1/3, 1]$, with $Q = 1/3$ at degeneracy and $Q \rightarrow 1$ at extreme hierarchy. As a structureless null hypothesis we draw mass triples independently log-uniformly over the observed span $[m_e, m_\tau]$ (2×10^6 samples; Appendix A). The resulting ensemble has $\langle Q \rangle = 0.525$, median 0.489, standard deviation 0.137, and

$$\Pr(|Q - \frac{2}{3}| \leq \sigma_{Q_{\text{exp}}}) \approx 1.1 \times 10^{-5}. \quad (23)$$

The i.i.d. log-uniform ensemble is deliberately structureless: it does not condition on the observed mass hierarchy, and one may object that hierarchical spectra populate the small- $|Q - 2/3|$ region differently. We therefore

also quote a *correlated (hierarchical) null* that enforces the hierarchy by construction. Procedure, fully specified for reproduction: (i) fix $m_3 = m_\tau$; (ii) draw the two log-gaps $g_{32} = \log_{10}(m_3/m_2)$ and $g_{21} = \log_{10}(m_2/m_1)$ independently and uniformly on $[1.0, 3.0]$ —a decade window that brackets the observed values $g_{32} = 1.226$ and $g_{21} = 2.315$ while making every draw strictly hierarchical; (iii) set $m_2 = m_3 10^{-g_{32}}$ and $m_1 = m_2 10^{-g_{21}}$; (iv) generate 2×10^6 triples with seed 43 (NumPy `default_rng`), compute Q for each, and report the fraction with $|Q - \frac{2}{3}| \leq \sigma_{Q_{\text{exp}}}$. This yields

$$\Pr(|Q - \frac{2}{3}| \leq \sigma_{Q_{\text{exp}}} \mid \text{hierarchical null}) = [P_{\text{corr}} = \dots],$$

to be compared directly with Eq. (23). **[Authors: run the four-step procedure above and insert the number; deposit the script alongside Appendix A item (iv), of which it is a strict variant.]** A coincidence probability of comparable smallness under both nulls would show that the significance of Eq. (20) is not an artifact of the structureless sampling assumption; a materially larger P_{corr} would attribute part of the apparent precision to the hierarchy itself, and the language of this subsection should then be tempered accordingly.

Under the structureless null, agreement with $2/3$ at the current experimental precision is a coincidence at the 10^{-5} level. This quantifies—without replacing—the usual qualitative statement that Eq. (1) is “surprisingly” precise. We emphasize what Eq. (23) does and does not establish: it disfavors the structureless null against the circulant mechanism, but the values $\kappa = 1/\sqrt{2}$ and $\delta = 2/9$ are recognized in the data rather than derived from a deeper principle; the null is disfavored, not excluded.

D. Radiative stability

The masses in Eq. (1) are pole masses, and one may worry that an exact relation among them cannot be fundamental because running masses at a common scale violate it at the percent level. Sumino has shown that a family gauge symmetry can cancel the QED radiative correction to Eq. (1), rendering the pole-mass relation exact in a well-defined effective theory [21, 22]. Our Proposition 1 is agnostic to this ultraviolet completion: it characterizes the algebraic structure any such completion must deliver at low energy—a Z_3 -circulant coupling operator at spectral equipartition.

V. PROPOSED EXPERIMENTAL TESTS

We organize the experimental program in three tiers. Tier A tests the spectral identity of Sec. IV; Tier B tests the confinement/internal-clock mechanism of Sec. II; Tier C comprises two laboratory configurations that probe the *motivating* vacuum-structure framework rather

than the theorems themselves—we state this explicitly, and note that for Tier C the standard-theory prediction is null, so that either outcome is informative: a positive signal is new physics, a null result sets the first quantitative bounds on the couplings in question. Each test is given with its method, present status, and decision criterion.

A. Tier A: tests of the spectral identity

E1. Precision of the τ mass (now–2030). The uncertainties on both Q_{exp} and the fitted phase δ are dominated entirely by Δm_τ ; both scale linearly with it. The most precise single measurement is from Belle II, $m_\tau = 1777.09 \pm 0.14$ MeV [28]; the current world average including it is $m_\tau = 1776.93 \pm 0.09$ MeV [23]. With the anticipated Belle II improvement to $\Delta m_\tau \simeq 0.03$ MeV, the propagated uncertainties become $\sigma_Q \simeq 1.7 \times 10^{-6}$ and $\sigma_\delta \simeq 2.1 \times 10^{-6}$ —a threefold sharpening of both rational-value tests. *Decision criterion:* if the central values continue to satisfy $|Q - 2/3| \lesssim \sigma_Q$ and $|\delta - 2/9| \lesssim \sigma_\delta$ as the errors shrink, the equipartition mechanism is progressively confirmed; a drift of either quantity away from its rational value at $\gtrsim 3\sigma$ falsifies it. This is the sharpest near-term test and requires no new experimental technique—only the already-planned Belle II dataset.

E2. Fourth-generation exclusion: $Q_4 = 1/2$ as a standing prediction. For n generations coupled through a Z_n -circulant at equipartition, the same two trigonometric sums give

$$Q_n = \frac{1 + 2\kappa^2}{n} \xrightarrow{\kappa^2=1/2} \frac{2}{n}, \quad (24)$$

with the observed $n = 3$ value $Q_3 = 2/3$. A sequential fourth charged lepton ℓ_4 would have to complete the spectrum to $Q_4 = (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau} + \sqrt{m_{\ell_4}})^{-2} \sum_k m_k = 1/2$. This condition is quadratic in $\sqrt{m_{\ell_4}}$ and has exactly two roots. With the PDG masses they are

$$\begin{aligned} m_{\ell_4} &= 95.1134 \pm 0.0103 \text{ MeV} \quad \text{and} \\ m_{\ell_4} &= 9.3204 \pm 0.0003 \text{ GeV}, \end{aligned}$$

where the uncertainties are propagated from $\Delta m_\tau = 0.09$ MeV (both central values shift by well under 10^{-3} per σ). Both roots are excluded, and have been for decades: a charged lepton at 95 MeV—below the muon mass—could not have escaped detection in any era of particle physics since the muon itself, and LEP2 excludes sequential charged leptons below ≈ 100 GeV—an order of magnitude above the second root, and three orders of magnitude above the first. The mechanism therefore does not merely permit a test at future colliders; it makes an unconditional standing prediction: *no sequential fourth charged lepton exists*. Discovery of one, at any mass, would falsify Z_n equipartition outright, since neither allowed root survives. No future facility is required, and

every future charged-lepton search is a continuing test of the mechanism.

An abstract-ready statement: “With the measured lepton masses, $Q_4 = 1/2$ has exactly two roots, $m_{\ell_4} = 95.1$ MeV and 9.32 GeV, both long excluded; the framework thus predicts unconditionally that no sequential fourth charged lepton exists.”

E3. Quark-sector circulant breaking (executed). We have carried out the calculation defined in the previous version of this test; the outcome is reported here as a result, not a proposal. The six quark masses were run with four-loop $\overline{\text{MS}}$ β and γ_m functions and tree-level continuous flavor matching (the four-loop treatment is required: strict two-loop running misses the $m_c(m_Z) \simeq 0.62$ GeV benchmark by +6.2%, whereas the treatment used here gives +2.1% for $m_c(m_Z)$ and +0.2% for $m_b(m_Z)$), and each triplet was fit to the circulant form (16) at $\mu = 2, 10, 91.2$, and 1000 GeV. The first finding is structural and exact: *pure QCD cannot generate any scale dependence of the circulant parameters.* The $\overline{\text{MS}}$ mass anomalous dimension γ_m is flavor-universal, so all masses within a sector rescale by a common factor between any two scales, and tree-level matching preserves mass ratios across thresholds; since κ , δ , and Q depend only on mass ratios, they are exactly RG-invariant under QCD (the numerical spread across the four scales is $< 10^{-14}$, i.e., integrator noise). The scale-tracking pattern conjectured in the earlier version of this test—a single breaking parameter running with α_s —therefore cannot exist in pure QCD, independently of any fit. The fits quantify what remains. With PDG-2024 inputs and Monte Carlo error propagation (2000 draws), the fitted amplitudes are $\kappa_{\text{up}} = 0.9114 \pm 0.0002$ and $\kappa_{\text{down}} = 0.7880 \pm 0.0007$. Both exceed the equipartition value $1/\sqrt{2} = 0.7071$ —the breaking has the same sign in both sectors, the one qualitative feature consistent with a common origin. But no single parameter covers both sectors: the excesses $\Delta\kappa = \kappa - 1/\sqrt{2}$ are 0.2043 (up) and 0.0809 (down), a ratio of 2.53, and the two κ values differ by $\sim 180\sigma$; a common no-intercept fit leaves a maximum residual comparable to the signal itself ($\max|\text{res}|/\Delta\kappa = 1.01$). The phases are equally far from the lepton pattern: $\delta_{\text{up}} = 0.0521 \pm 0.0001$ and $\delta_{\text{down}} = 0.1005 \pm 0.0006$, neither near $2/9 = 0.2222$; correspondingly $Q_{\text{up}} = 0.887$ and $Q_{\text{down}} = 0.747$ against the lepton value $2/3$. The consequence is the one this test was designed to force: *as far as this test sees, spectral equipartition is confined to the charged-lepton sector.* Any extension of the mechanism to quarks cannot be driven by QCD running and would require a non-QCD driver—electroweak/Yukawa contributions to the running, which are neglected and uncomputed here, and about whose magnitude we make no claim. The same-sign excess of both κ ’s above $1/\sqrt{2}$ is recorded as an unexplained regularity, not as support.

E4. Neutrino null test (KATRIN/Project 8 + cosmology). If the equipartition condition is specific to the charged-lepton coupling channel, the neutrino Koide ratio Q_ν should be statistically unremarkable, i.e., con-

sistent with the hierarchical null of Sec. IV C, whereas a universal mechanism would enforce $Q_\nu = 2/3$. Oscillation data fix the two mass-squared differences; one absolute-scale measurement closes the system. Current direct limits ($m_\beta < 0.8$ eV [29]) and cosmological bounds on $\sum m_\nu$ already restrict Q_ν to a computable band for each ordering; a Project-8-class measurement or a robust cosmological detection of $\sum m_\nu$ converts this into a sharp discrimination between channel-dependent and universal equipartition.

B. Tier B: tests of the confinement/internal-clock mechanism

E5. Clock dilation of composite systems. The prediction of Sec. IID—that a confined object’s internal mode defines its proper time and dilates kinematically by γ —is already verified to high precision for one internal clock, the muon lifetime in storage rings [30], and the operational identification of mass with clock frequency has been demonstrated interferometrically [24]. The open extension is to systems with a *resolvable structural* mode: Compton-clock interferometry performed on composite particles (molecules, clusters) in states of different internal excitation. The mechanism of Sec. IID requires the clock rate to track the confined-structure deformation exactly as Eq. (8) maps the rest configuration—i.e., no anomalous dependence of dilation on internal-mode occupation. Any such dependence at current interferometric sensitivity would falsify the identification.

E6. The hadronic breathing mode. The ϕ^4 kink’s shape mode $\omega_1 = (\sqrt{3}/2)m$ has a physical counterpart: in topological-soliton descriptions of the nucleon, the Roper resonance $N(1440)$ is interpreted as the breathing mode of the baryonic field configuration [13, 31]. The framework-level statement is that the ratio of internal-mode frequency to constituent-field mass scale is fixed by the confining structure, not adjustable per particle. A systematic confrontation—breathing-mode frequencies across the baryon octet from lattice QCD versus the soliton prediction pattern—is a defined test of whether physical hadrons carry the internal-clock structure that Sec. IID exhibits exactly in 1+1 dimensions.

C. Tier C: laboratory probes of vacuum–topology coupling

The two configurations below test whether vacuum fluctuation structure couples to vector-potential *topology* in regions where the magnetic field itself vanishes. Standard Lifshitz theory predicts null results in both; they are included because they are inexpensive, sharply defined, and bound-setting regardless of outcome. Both build on the recent demonstration that Casimir forces in ferrofluid media are magnetically tunable and measurable with a purpose-built cantilever [32], and on the theoret-

ical demonstration that rotation confined to a restricted spatial domain imprints a geometric (Aharonov–Bohm-like) phase on quantum-vacuum interactions outside it [33].

E7. Static test: Casimir force inside a Halbach toroidal bore. A Halbach ring of N permanent magnets produces a strong field confined to its bore region with near-zero exterior field; conversely, a toroidal winding confines $\mathbf{B}$ to the torus body, leaving the bore with $\mathbf{B} \approx 0$ but nontrivial vector potential $\mathbf{A}$. The proposed measurement compares the sphere–plate Casimir force in ferrofluid [32] in three configurations: (i) inside the bore ($\mathbf{B} \approx 0$, $\mathbf{A} \neq 0$, nontrivial winding topology); (ii) outside the assembly at matched residual field ($\mathbf{B} \approx 0$, $\mathbf{A} \approx 0$); (iii) assembly removed (control). Lifshitz theory with the measured local $\mathbf{B}$ predicts configurations (i)–(iii) to agree within the ferrofluid’s response to field leakage, which is independently mapped with a Hall probe and corrected. *Decision criterion:* a reproducible force difference between (i) and (iii) surviving the leakage correction, absent between (ii) and (iii), would indicate coupling of the fluctuation spectrum to the potential’s topology; a null result bounds any such coupling at the fractional-force sensitivity of the apparatus ($\sim 10^{-2}$ at 100 nm separations with the technique of Ref. [32]).

E8. Dynamic test: counter-rotating toroidal magnets with 2Ω lock-in accelerometry. Two coaxial toroidal (Halbach) assemblies rotating at $+\Omega$ and $-\Omega$ produce, in the annular border region between them, superposed counter-rotating vector potentials whose beat oscillates at exactly 2Ω while the local $\mathbf{B}$ remains near zero; on the symmetry axis, $\mathbf{A}$ vanishes identically, furnishing an internal null channel. The measurement is a lock-in accelerometer search at reference frequency 2Ω (derived from the shaft encoder) at three stations: border region, bore, and axis. Mechanical systematics are suppressed by the counter-rotation symmetry (motor vibrations at Ω cancel in the horizontal plane) and diagnosed by three discriminators: the genuine channel must appear in the border region but not on the axis, reverse phase under reversal of one rotor, and vanish when the magnet assemblies are replaced by non-magnetic dummies of equal inertia. Standard theory predicts a null; the established adjacent effect—the quantum-vacuum Sagnac phase of a rotating body [33]—is many orders below MEMS sensitivity. The experiment therefore sets the first laboratory upper bounds on anomalous inertial or gravitational coupling to *rotating* vector-potential topology, with a clean upgrade path (MEMS $\rightarrow$ seismometer $\rightarrow$ atom interferometer) improving the bound by $\sim 10^8$ without changing the principle.

D. Decision structure

The tiers are logically independent: E1–E4 can confirm or falsify the spectral mechanism irrespective of Tier C outcomes, and null results in E7–E8 constrain only

the motivating framework, not Propositions of Secs. II–IV. The near-term critical path is E1 (data already accumulating) and E7 (single-laboratory tabletop scale); E3 has been executed (Sec. V A), with its outcome—equipartition confined to the lepton sector, and no possible QCD-driven scale dependence—incorporated above. For completeness we note that the broader framework also implies a cosmological discriminator—a Milgromian acceleration scale tracking the Hubble rate, $a_0(z) = cH(z)/2\pi$ [34, 35], testable through the redshift evolution of the baryonic Tully–Fisher normalization—which lies outside the claims of this paper and is developed elsewhere.

VI. DISCUSSION: STATUS OF EACH CLAIM

It is essential to state plainly what has and has not been established.

Established as mechanism (machine-verified). (i) In the model of Eq. (2), inertial mass is trapped field energy: $E = \gamma E_0$, $P = \gamma E_0 u$, $F = E_0 a$ [Eqs. (9)–(11)], with no kink-mass parameter in the theory. (ii) Inertial and gravitational mass are the same integral [Eq. (12)]—an identity inherited from minimal coupling, realized by the soliton with no free mass parameter (the 1+1D caveat of Sec. II C applies). (iii) The soliton carries an internal clock at $\omega_1 = (\sqrt{3}/2)m$ whose boost dilation is derived [Sec. II D]. (iv) $Q = 2/3$ is an exact identity of Z_3 -circulant coupling at equipartition, and only there [Proposition 1].

Consistent with data at current precision. $Q_{\text{exp}} = 2/3$ at 0.4σ ; $\delta = 2/9$ at 0.4σ ; coincidence probability under the structureless null $\approx 1.1 \times 10^{-5}$. Two further results are now in hand. First, the $Q_4 = 1/2$ condition has exactly two roots with the measured masses, 95.1134 ± 0.0103 MeV and 9.3204 ± 0.0003 GeV, both long excluded—the framework’s prediction that no sequential fourth charged lepton exists is unconditional and is currently satisfied. Second, the executed quark-sector analysis (E3) shows that the circulant parameters are exactly RG-invariant under pure QCD and that no single breaking parameter covers both quark sectors ($\kappa_{\text{up}} = 0.9114(2)$ versus $\kappa_{\text{down}} = 0.7880(7)$, a $\sim 180\sigma$ separation); equipartition is, as far as that test sees, a lepton-sector fact, with the same-sign excess of both quark κ ’s above $1/\sqrt{2}$ recorded as an unexplained regularity.

Reinterpretation of known physics. The reading of a particle’s proper time as internal-mode phase, and of the photon’s vanishing proper time as the absence of trapped structure, reorganizes standard relativity around a confinement ontology; it adds no new equation at this stage, though it motivates tests E5 and E6 of Sec. V.

Open, with defined calculations. (1) Derive $\kappa = 1/\sqrt{2}$ from a variational or information-theoretic principle on the coupling operator, rather than recognizing it in data. (2) Derive $\delta = 2/9$, or demote it to numerology; its radiative stability within the Sumino mechanism [22]

constrains candidate derivations. (3) Identify the non-QCD driver—electroweak/Yukawa running, uncomputed here—that any quark-sector extension of equipartition now requires, given that QCD running is exactly inert on the circulant parameters (Sec. V A, E3); absent such a driver, explain why the mechanism selects the charged leptons. (4) Promote the confinement mechanism to 3+1 dimensions explicitly within the Skyrme/QCD setting of Sec. III, connecting ω_1 -type internal modes to physical hadronic excitations. The broader program in which these results are embedded—spacetime and gravitation as emergent from more primitive structure—connects to established lines of research [25–27] but is not advanced by the present paper and is deliberately excluded from its claims.

The referee-style demand that a framework of this kind produce “one number that lands on a measurement” is met by Eq. (21): a two-parameter reconstruction of three masses spanning four orders of magnitude, with both dimensionless constants of the reconstruction (κ^2 , δ) consistent with simple rationals ($1/2$, $2/9$) within propagated experimental errors, and a coincidence probability of order 10^{-5} under the structureless alternative. Whether these rationals are fundamental is precisely what calculations (1)–(2) and the $Q_4 = 1/2$ test will decide.

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Appendix A: Verification and numerical methods

Symbolic verification. All identities of Secs. II and IV were verified in SymPy: substitution of Eq. (5) and Eq. (8) into Eq. (3) (residual 0); the Bogomolny identity (residual 0); Eq. (6) by direct integration; Eqs. (9) and (10) via the substitution $\xi = \gamma(x - ut)$ (residuals 0); the series Eq. (11); and the sums leading to Eq. (18) (exact).

Numerical checks. (i) Boosted-kink energy: adaptive quadrature of T^{00} at $(\lambda, v, u) = (2, 1, 0.6)$ gives $E = 1.666667 = \gamma E_0$. (ii) Fluctuation spectrum: Eq. (13) discretized by second-order finite differences on $x \in [-20, 20]$ with $N = 8000$ points and diagonalized as a symmetric tridiagonal system yields $\omega^2 = \{-0.00000, 2.99999\}$ discrete and continuum on-set $4.007 \approx m^2 = 4$, i.e., $\omega_1/m = 0.866024$ vs. $\sqrt{3}/2 = 0.866025$. (iii) Mass fit: Levenberg–Marquardt minimization of Eq. (16) against $(\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu})$ with $\kappa = 1/\sqrt{2}$; uncertainties by refitting at $m_\tau \pm \Delta m_\tau$. (iv) Null ensemble: 2×10^6 i.i.d. triples with $\log_{10} m$ uniform on $[\log_{10} m_e, \log_{10} m_\tau]$, seed 42; Eq. (23) is the empirical fraction. All scripts are short (under one page each) and available from the author.

Appendix B: Change log (v2 $\rightarrow$ v3)

- **E-1.** Split recommendation (Secs. II–III as a pedagogy paper; Secs. IV–V as the publishable kernel) recorded as submission strategy; no in-text change—the single-paper form is retained for this cycle with the minimum register repairs E-3 and E-4 applied.
- **E-2.** Test E2 (Sec. V A) rewritten: both roots of $Q_4 = 1/2$ computed (95.1134 ± 0.0103 MeV and 9.3204 ± 0.0003 GeV; values as updated by E-10), both long excluded—E2 converted from a conditional collider test into an unconditional standing prediction that no sequential fourth charged lepton exists.
- **E-3.** Sec. II C register fix: $m_{\text{grav}} = m_{\text{inert}}$ restated as an identity inherited from minimal coupling (exhibited, not derived anew), with the mandatory 1+1D caveat; companion change to Sec. VI item (ii).
- **E-4.** Sec. II D: $\omega_1 = (\sqrt{3}/2)m$ stated as exact (Pöschl–Teller closed-form spectrum); the $N = 8000$ finite-difference diagonalization reclassified as confirmation.
- **E-5.** Unreplicated Tajmar reference (old [34]) deleted together with its E8 clause; references renumbered consecutively ([35] $\rightarrow$ [34] Milgrom, [36] $\rightarrow$ [35] McGaugh); the Sec. V D citation updated from [35, 36] to [34, 35].
- **E-6.** Sec. IV C: correlated/hierarchical null added immediately after Eq. (23) with a fully specified four-step procedure and a clearly marked placeholder [$P_{\text{corr}} = \dots$]; the retained closing sentence reworded from “Under a featureless hierarchical null” to “Under the structureless null”.
- **E-7.** Test E3 (Sec. V A) rewritten as an executed result: circulant parameters exactly RG-invariant under pure QCD ($< 10^{-14}$ spread over $\mu = 2$ –1000 GeV); $\kappa_{\text{up}} = 0.9114(2)$, $\kappa_{\text{down}} = 0.7880(7)$; equipartition confined to the lepton sector; companion update to the Sec. V D critical-path sentence.
- **E-8.** Sec. IV B: sentence added fixing the δ -fit convention at the 10^{-6} level (convention spread 4.6×10^{-6} ; values as updated by E-10), naming the Appendix A convention (absolute least-squares with κ fixed) as the one producing Eq. (21).
- **E-9.** Abstract and Sec. VI (“Consistent with data at current precision”; “Open, with defined calculations”) updated to carry E-2 and E-7 as results rather than proposals.

- **E-10.** τ -mass correction (abstract; Secs. IV B, V A, VI; Table I; test E2). v2 quoted $m_\tau = 1776.86 \pm 0.12$ MeV and attributed it to the Belle II measurement [28]; that measurement is in fact $m_\tau = 1777.09 \pm 0.14$ MeV, and 1776.86 ± 0.12 MeV was the pre-Belle-II world average. v3 uses the current world average including Belle II, $m_\tau = 1776.93 \pm 0.09$ MeV [23] (pdgLive node S035, Re-

view of Particle Physics, 2026 edition). All dependent quantities updated: Eqs. (20), (21), and (23), Table I, the δ -convention values, the E1 sharpening factor, and the $Q_4 = 1/2$ roots. Net effect: agreement with the rational values tightens from 0.9σ to $\approx 0.4\sigma$ for both $Q = 2/3$ and $\delta = 2/9$.

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